

25 微分.

1. $\sum_{n=2}^{\infty} (-1)^n \frac{n^4}{2^n}$ 敛散性

$$a_n = \frac{n^4}{2^n} \quad \lim_{n \rightarrow \infty} a_n = 0. \quad \text{convergent.}$$

2. $\sum_{n=1}^{\infty} (1 - \frac{\ln n}{n})^n$ 敛散性.

$$a_n = (1 - \frac{\ln n}{n})^n.$$

$$\lim_{n \rightarrow \infty} a_n = (1 + (-\frac{\ln n}{n}))^{(-\frac{n}{\ln n})(-\frac{\ln n}{n}) \cdot n} = \lim_{n \rightarrow \infty} e^{-\ln n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0. \quad \text{失败}$$

Another Way,

methods: 积分判别法 / Limit Comparison Test / Ratio/Root Test.

Try Ratio/Root Test.

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = (1 - \frac{\ln(n+1)}{n+1})^{n+1} / (1 - \frac{\ln n}{n})^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} (1 - \frac{\ln n}{n}) = 1 \quad \text{失败.}$$

Only. Comparison Test $e \cdot (1 - \frac{\ln n}{n})^n = \frac{1}{n}$

$$\ln(H^X) \sim X.$$

$$\lim_{n \rightarrow \infty} \frac{(1 - \frac{\ln n}{n})^n}{\frac{1}{n}} = \lim_{n \rightarrow \infty} n (1 - \frac{\ln n}{n})^n = \lim_{n \rightarrow \infty} e^{\ln n + n \ln(1 - \frac{\ln n}{n})} = \lim_{n \rightarrow \infty} e^{\ln n + n \cdot (-\frac{\ln n}{n})}$$

$$\ln n + n \ln \frac{n - \ln n}{n} = \ln n + n \ln(n - \ln n) - n \ln n$$

$$= e^0 = 1.$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverge}$$

$$\therefore \sum_{n=1}^{\infty} (1 - \frac{\ln n}{n})^n \text{ diverge.}$$



3. $r = 4(1 + \sin\theta)$ 求 $\theta = \frac{\pi}{2}$ 曲率. (注: θ 应为 $-\frac{\pi}{2}$, 不知是否抄录有误).

曲率必背

$$K = \frac{|y''|}{(1+y'^2)^{\frac{3}{2}}} = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{\frac{3}{2}}}$$

(r, θ) 变. $r \rightarrow r(\theta)$.

$$x = r \cos \theta, y = r \sin \theta.$$

$$x' = \frac{d}{d\theta} (r \cos \theta) = r' \cos \theta - r \sin \theta.$$

$$y' = r' \sin \theta + r \cos \theta.$$

$$x'' = r'' \cos \theta - r' \sin \theta - r' \sin \theta - r \cos \theta.$$

$$= r'' \cos \theta - 2r' \sin \theta - r \cos \theta$$

$$y'' = r'' \sin \theta + r' \cos \theta + r' \cos \theta - r \sin \theta.$$

$$= r'' \sin \theta + 2r' \cos \theta - r \sin \theta.$$

$$(x'^2 + y'^2)^{\frac{3}{2}} = (r'^2 \cos^2 \theta - 2r'r' \sin \theta \cos \theta + r^2 \sin^2 \theta + r'^2 \sin^2 \theta + r^2 \cos^2 \theta + 2r'r \sin \theta \cos \theta)^{\frac{3}{2}} \\ = (r'^2 + r^2)^{\frac{3}{2}}$$

$$|x'y'' - y'x''| = \left| \begin{array}{l} r^2 r'' \cos^2 \theta \sin \theta - r^2 r'' \sin^2 \theta + 2r'^2 \cos^2 \theta - 2r'r' \sin \theta \cos \theta - r'r \sin \theta \cos \theta \\ + r^2 \sin^2 \theta \end{array} \right| \\ + \dots$$

$$= \left| r^2 + 2(r')^2 - rr'' \right|$$

$$\therefore K = \frac{|r^2 + 2r'^2 - rr''|}{(r^2 + r'^2)^{\frac{3}{2}}} = \frac{|64 + 32|}{8^3} = \frac{96}{512} = \frac{3}{16}.$$

$$r = 4(1 + \sin \theta) = 8.$$

$$r' = 4 \cos \theta = 0.$$

$$r'' = -4 \sin \theta = -4.$$



4/. $x^2 + y^2 + 2z = 2$ 在 $(1, 2, 3)$ 的切平面.

Review: 切线与法平面.

曲线 $\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$ 切线 $\frac{x-x_0}{x'(t_0)} = \frac{y-y_0}{y'(t_0)} = \frac{z-z_0}{z'(t_0)}$. $\langle x'(t_0), y'(t_0), z'(t_0) \rangle$ 切向量.

法平面: $x'(t_0)(x-x_0) + y'(t_0)(y-y_0) + z'(t_0)(z-z_0) = 0$.

切平面与法线.

切平面: $F_x(x-x_0) + F_y(y-y_0) + F_z(z-z_0) = 0$.

法线: $\frac{x-x_0}{F_x} = \frac{y-y_0}{F_y} = \frac{z-z_0}{F_z}$.

$F(x(t), y(t), z(t)) \equiv 0$.

$\vec{n} = \langle F_x, F_y, F_z \rangle$.

$F = x^2 + y^2 + 2z - 2$.

$F_x = 2x = 2$, $F_y = 2y = 4$, $F_z = 2z = 12$

$2(x-1) + 4(y-2) + 12(z-3) = 0$

$x + 2y + 6z = 23$

5/.

$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3y + xy^4 + x^2y}{x+y} \quad \exists \text{ or } \nexists$

Analyse: $x=y=0 \rightarrow 0$. $\frac{\text{分子} \rightarrow \text{高阶无穷小量}}{\text{分母}}$

way: 把分母也变高阶无穷小量! $\rightarrow$

$y = -x + x^3$

分母 $= x^3$

分子中最低阶 $x^2y = x^2(-x + x^3) \Rightarrow -x^3$

$\therefore \lim = -1$

NOT $\exists$!

#.



6/. Find extreme value of $z = \cos^2 x + \cos^2 y$ subject to the constraint $x+y = \frac{\pi}{4}$

$$L = \cos^2 x + \cos^2 y + \lambda (x+y - \frac{\pi}{4})$$

$$\begin{cases} L_x = \lambda + (-\sin x) \cdot 2 \cos x = 0 \\ L_y = \lambda + (-\sin y) \cdot 2 \cos y = 0 \\ L_\lambda = x+y - \frac{\pi}{4} = 0 \end{cases} \Rightarrow$$

$$y \in [0, 2\pi] \cdot \begin{cases} x = \frac{\pi}{8} & x = -\frac{3\pi}{8} \\ y = \frac{\pi}{8} & y = \frac{5\pi}{8} \end{cases}$$

$$1^\circ \quad x = \frac{\pi}{8} \quad y = \frac{\pi}{8} \quad \cos^2 \frac{\pi}{8} = \frac{2+\sqrt{2}}{4}$$

$$z = \frac{2+\sqrt{2}}{2}$$

$$2^\circ \quad x = \frac{3\pi}{8} \quad y = \frac{5\pi}{8} \quad \cos^2 \left(\frac{3\pi}{8}\right)^2 = \frac{2-\sqrt{2}}{4}$$

$$z = \frac{2-\sqrt{2}}{2}$$

$$\therefore z_{max} = \frac{2+\sqrt{2}}{2}$$

$$z_{min} = \frac{2-\sqrt{2}}{2}$$

7/.

$$\sum_{n=1}^{\infty} (-1)^n \frac{(2x-3)^n}{n \cdot 4^n} \quad \text{收敛半径. 和函数.}$$

$$(-1)^n \cdot \frac{1}{n} \cdot \left(\frac{2x-3}{4}\right)^n$$

$$a_n = \frac{(-1)^n}{n}, \quad t = \frac{2x-3}{4}, \quad \sum_{n=1}^{\infty} a_n t^n$$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$$

$$1^\circ \quad t=1, \quad x = \frac{7}{2}, \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{n} \quad \text{converge.} \quad \therefore \text{C.S. } \left(-\frac{1}{2}, \frac{7}{2}\right]$$

$$2^\circ \quad t=-1, \quad x = -\frac{1}{2}, \quad \sum_{n=1}^{\infty} \frac{1}{n} \quad \text{diverge}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$\ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$$

$$-\ln(1+t) = -t + \frac{t^2}{2} - \frac{t^3}{3} + \dots = \sum_{n=1}^{\infty} a_n t^n$$

$$\therefore = -\ln\left(\frac{2x+1}{4}\right), \quad x \in \left(-\frac{1}{2}, \frac{7}{2}\right]$$



8/. $F(x + \frac{z}{y}, y + \frac{z}{x}) = 0$ 确定隐函数 $z = f(x, y)$. 求证:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - xy.$$

$$u = x + \frac{z(x, y)}{y}, v = y + \frac{z(x, y)}{x}$$

$$F(u, v)$$

$$u, v$$

$$x: F_u \frac{\partial u}{\partial x} + F_v \frac{\partial v}{\partial x} = F_u \cdot (1 + \frac{z_x}{y}) + F_v \cdot (\frac{xz_x - z}{x^2}) = 0 \quad (1)$$

$$y: F_u \frac{\partial u}{\partial y} + F_v \frac{\partial v}{\partial y} = F_u \cdot (\frac{yz_y - z}{y^2}) + F_v \cdot (1 + \frac{z_y}{x}) = 0 \quad (2)$$

$$\text{From (1). } F_u = F_v \left(\frac{xz_x - z}{x^2} \right) \frac{-1}{1 + \frac{z_x}{y}} \quad \dots (3)$$

$$(3) \rightarrow (2): F_v \left(\frac{xz_x - z}{x^2} \right) \frac{-1}{1 + \frac{z_x}{y}} \left(\frac{yz_y - z}{y^2} \right) + F_v \left(1 + \frac{z_y}{x} \right) = 0.$$

$$-(xz_x - z)(yz_y - z) + (x^2 + xz_y)(y + yz_x) = 0.$$

$$xy(x + z_y)(y + z_x).$$

$$(xy + yz_y)(xy + xz_x)$$

$$A = x \frac{\partial z}{\partial x}, B = y \frac{\partial z}{\partial y}, C = z, D = xy$$

$$\text{要证 } A + B = C - D.$$

$$AB - BC - AC + C^2$$

$$\text{有 } -(A - C)(B - C) + (D + A)(D + B) = 0.$$

$$-AB + BC + AC - C^2 + D^2 + AD + BD + AB = 0.$$

$$C(A + B) + D(A + B) - (C^2 - D^2)$$

$$(A + B)(D + C) + (D + C)(D - C) = 0$$

$$(D + C)(A + B + D - C) = 0$$

0

$$\therefore A + B + D - C = 0$$

$$\therefore A + B = C - D \quad \#.$$

编者按:

$$\text{用 } z_x = -\frac{G_x}{G_z}$$

$$z_y = -\frac{G_y}{G_z}$$

$$G(x, y, z)$$

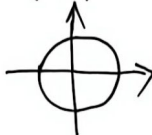
$$= F(u, v).$$

会更 easy



$$9/. \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (x^2+y^2)^{\frac{3}{2}} dy dx.$$

$$-\sqrt{9-x^2} \leq y \leq \sqrt{9-x^2} \quad -3 \leq x \leq 3.$$

$$y = -\sqrt{9-x^2} \quad y^2 \leq 9-x^2.$$


$$= \int_0^{2\pi} d\theta \int_0^3 r^3 r dr.$$

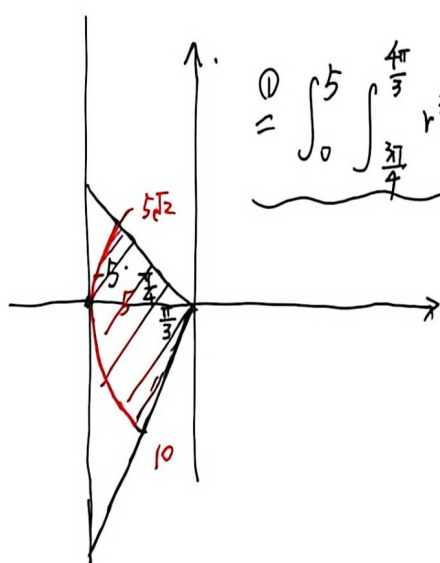
$$= 2\pi \cdot \frac{3^5}{5} = 2\pi \cdot \frac{81 \times 3}{5} = \frac{486\pi}{5}.$$

$$10/. \int_{\frac{3\pi}{4}}^{\frac{4\pi}{3}} d\theta \int_0^{-5\sec\theta} r^3 \sin\theta dr. \quad (\text{交换积分次序, 不用计算}).$$

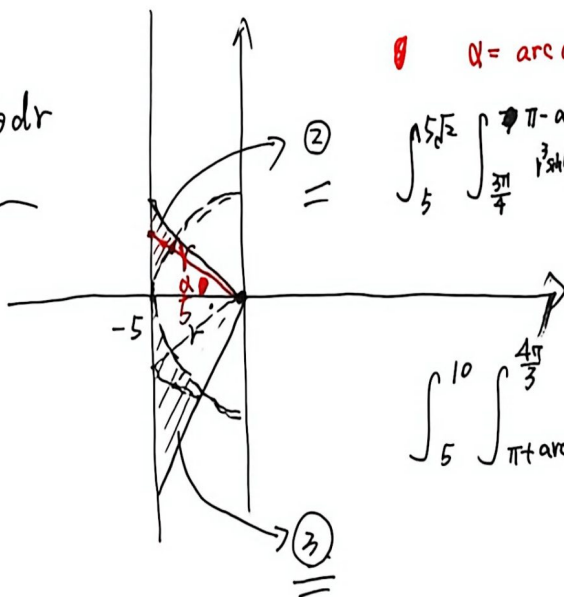
$$\frac{3\pi}{4} < \theta < \frac{4\pi}{3}, \quad 0 < r < -\frac{5}{\cos\theta} = -\frac{5}{\sec\theta}.$$

$$\downarrow$$

$$r = -5\sec\theta \quad r\cos\theta = -5$$



$$\textcircled{1} \int_0^5 \int_{\frac{3\pi}{4}}^{\frac{4\pi}{3}} r^3 \sin\theta d\theta dr$$



$$\theta = \arccos \frac{5}{r}.$$

$$\textcircled{2} \int_5^{5\sqrt{2}} \int_{\frac{3\pi}{4}}^{\pi - \arccos \frac{5}{r}} r^3 \sin\theta d\theta dr$$

$$\int_5^{10} \int_{\pi + \arccos \frac{5}{r}}^{\frac{4\pi}{3}} r^3 \sin\theta d\theta dr.$$

$$I = \textcircled{1} + \textcircled{2} + \textcircled{3}.$$



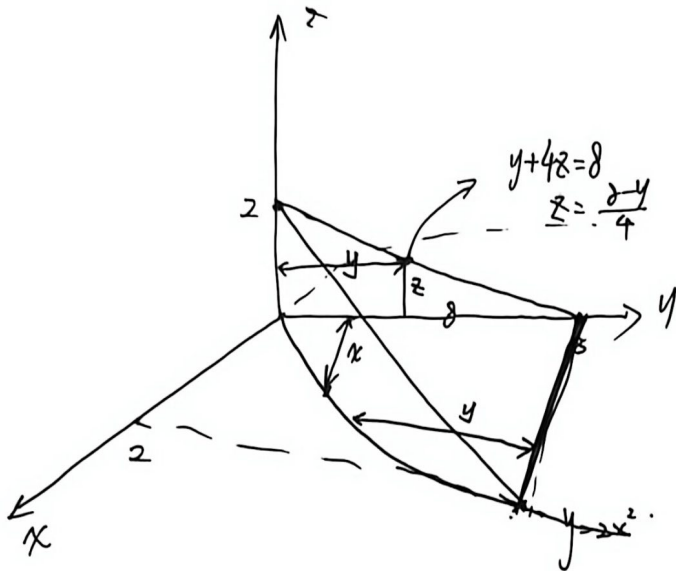
11/.
$$\int_0^1 \int_0^x \int_0^y \frac{\cos z}{(1-z)^2} dz dy dx.$$

$$\begin{cases} 0 \leq z \leq y \\ 0 \leq y \leq x \\ 0 \leq x \leq 1 \end{cases} \Rightarrow 0 \leq z \leq y \leq x \leq 1 \xrightarrow{\text{次序}} \begin{array}{l} \text{fix } z: 0 \leq z \leq 1, \\ y: z \leq y \leq 1 \\ x: y \leq x \leq 1. \end{array}$$

$$\begin{aligned} &\Rightarrow \int_0^1 \int_1^z \int_1^y \frac{\cos z}{(1-z)^2} dx dy dz. \\ &= \int_0^1 \frac{\cos z}{(1-z)^2} dz \int_1^z (y-1) dy \quad \xrightarrow{\frac{(z^2-1)}{2} - z + 1 = \frac{z^2 - 2z + 1}{2} = \frac{(z-1)^2}{2}} \\ &= \frac{1}{2} \int_0^1 \cos z dz = \underline{\underline{\frac{1}{2} \sinh 1}}. \end{aligned}$$



12/. Find volume of the solid in the first octant bdd by $y = 2x^2$, $y + 4z = 8$



$$0 \leq x \leq 2.$$

$$2x^2 \leq y \leq 8.$$

$$0 \leq z \leq \frac{8-y}{4}.$$

$$\int_0^2 dx \int_{2x^2}^8 dy \int_0^{\frac{8-y}{4}} dz = \int_0^2 dx \int_{2x^2}^8 \frac{8-y}{4} dy$$

$$= \int_0^2 2(8-2x^2) - \frac{64-4x^4}{8} dx$$

$$= \int_0^2 8 - 4x^2 + \frac{x^4}{2} dx \quad 16$$

$$= 16 - 4 \frac{8}{3} + \frac{32}{5} = \frac{240 - 32 \times 5 + 48}{15} = \frac{128}{15}.$$



13/.

$$y'' - 2y' + 2y = xe^{3x}.$$

$$r^2 - 2r + 2 = 0$$

$$r = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i \quad \alpha=1, \beta=1. \Rightarrow y = e^x (C_1 \cos x + C_2 \sin x).$$

$$y^* = x^0 \cdot e^{3x} \cdot (Ax+B) = e^{3x} Ax + Be^{3x}$$

$$\hookrightarrow (y^*)' = 3e^{3x}Ax + Ae^{3x} + 3Be^{3x}$$

$$(y^{*'})' = 9e^{3x}Ax + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x}.$$

$$9A - 6A + 2A = 1$$

$$9B - 6B + 2B + 6A - 2A = 0 \Rightarrow \begin{cases} A = \frac{1}{5} \\ B = -\frac{4}{5} \cdot \frac{1}{5} = -\frac{4}{25} \end{cases}$$

$$y^* = e^{3x} \left(\frac{1}{5}x - \frac{4}{25} \right).$$

$$Y = e^x (C_1 \cos x + C_2 \sin x) + e^{3x} \left(\frac{1}{5}x - \frac{4}{25} \right).$$

14/. $\iint_G (x^2 + z^2) dS$, G is the paraboloid $y = 1 - x^2 - z^2$ whose projection on to Xz -plane is $R = \{(x, z), x^2 + z^2 \leq 1\}$.

$$y_x = -2x, y_z = -2z.$$

$$\iint_G = \iint_R (x^2 + z^2) \sqrt{1+4x^2+4z^2} dx dz = \int_0^{2\pi} d\theta \int_0^1 r^2 \sqrt{1+4r^2} r dr = \int_0^{2\pi} d\theta \int_0^1 r^3 \sqrt{1+4r^2} dr.$$

$$\cancel{u = \sqrt{1+4r^2}} \quad \cancel{du = 8r dr} \quad u = 1+4r^2.$$

$$du = 8r dr.$$

$$r^2 = \frac{u-1}{4}.$$

$$I = \int_0^{2\pi} d\theta \int_1^5 \frac{u-1}{4} \cdot \sqrt{u} \cdot \frac{du}{8}$$

$$= \frac{1}{32} \int_0^{2\pi} d\theta \int_1^5 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du.$$

$$= 2\pi \cdot \frac{1}{32} \left(\frac{5^{\frac{5}{2}}-1}{\frac{5}{2}} - \frac{5^{\frac{3}{2}}-1}{\frac{3}{2}} \right).$$

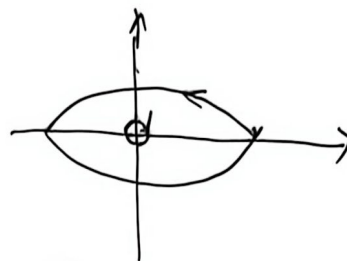
$$= \frac{\pi (25\sqrt{5}+1)}{60}$$



15/. $\oint_C \frac{(x-y)dx}{x^2+y^2} + \frac{(x+y)dy}{x^2+y^2}$ $C: \frac{x^2}{9} + \frac{y^2}{4} = 1$ positive direction.

P Q

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 \quad (0,0) \text{ No Def!}$$



$$\oint_C = \oint_{C+\epsilon^-} - \oint_{\epsilon^-} \quad \text{Let } \Gamma: x^2+y^2=\epsilon^2 \quad \underline{\epsilon \rightarrow 0} \quad \curvearrowright$$

$$= 0 - \oint_{\epsilon^-} = \oint_{\epsilon^+}$$

$$x = \epsilon \cos \theta, \quad y = \epsilon \sin \theta \quad dx = -\epsilon \sin \theta d\theta, \quad dy = \epsilon \cos \theta d\theta.$$

$$\oint_{\epsilon^+} \int_0^{2\pi} \frac{\epsilon(\cos \theta - \sin \theta)(-\epsilon \sin \theta)}{\epsilon^2} + \frac{\epsilon(\sin \theta + \cos \theta)\epsilon \cos \theta}{\epsilon^2} d\theta.$$

$$= \int_0^{2\pi} -\cos \theta \sin \theta + \sin^2 \theta + \sin \theta \cos \theta + \cos^2 \theta d\theta = 2\pi \quad \#1$$

16/. $\Omega: \text{bddd} \cdot x^2+y^2=4, z=0, z=3, \quad \vec{n} \text{ outer}$

$$\iint_{\partial \Omega} (x^3 + \tan y z) dy dz + (y^3 - e^{xz}) dz dx + (3z + x^3) dx dy.$$

Gauss. $\iiint_{\Omega} (3x^2 + 3y^2 + 3) dV$

$$= 3 \int_0^3 dz \int_0^{2\pi} d\theta \int_0^2 (r^2 + 1) r dr$$

$$= \underset{9}{3} \cdot \underset{18\pi}{3} \cdot \underset{6}{2\pi} \cdot \left(\frac{16}{4} + \frac{4}{2} \right) = 108\pi.$$

